

Circuits — Copper Tape Card: Light It Up!

Planrs Academy | Grade 1 | 30 Minutes | Beginner

Get ready to make your very own glowing card! We will learn how electricity travels and lights up an LED.



Story Time

The light made Grandma so happy — it was like holding a little star!

Let's Begin!

Today, you will make your own glowing card too!

Priya's Glowing Diwali Card

A magical story of light and love!

Priya wanted to make something special for her grandmother. She made a beautiful Diwali card that glows! When she pressed the card, a tiny light lit up like magic!



To Build

Copper tape, LED light, and
coin cell battery

To Decorate

Cardstock, scissors, crayons,
and stickers

Meet Our Materials

We will use these to make our glowing card!

Touch and explore each material! Feel the shiny
copper tape, look at the tiny LED legs, and see
how small the battery is. These will all work
together to make light!



What is Electricity?

Electricity is like a tiny road for energy!

It travels through wires to make lights glow bright. Just like water flows in a river, electricity flows to bring us light and power!



Power Source — Battery

The Energy Box!

A battery gives power to make things work!
It is like a little energy box that stores electricity inside.

The coin cell battery is small and round like a coin!
It powers the LED lights in Diwali diyas and fairy lights.



06

Conductor = Road for Electricity

- Copper tape lets electricity flow through
- Like a smooth road for tiny energy cars!
- Metals are good conductors



07

Insulator = Wall that Stops Electricity

- Paper and plastic block electricity
- Like a wall that stops the energy cars!
- Keeps us safe from electricity

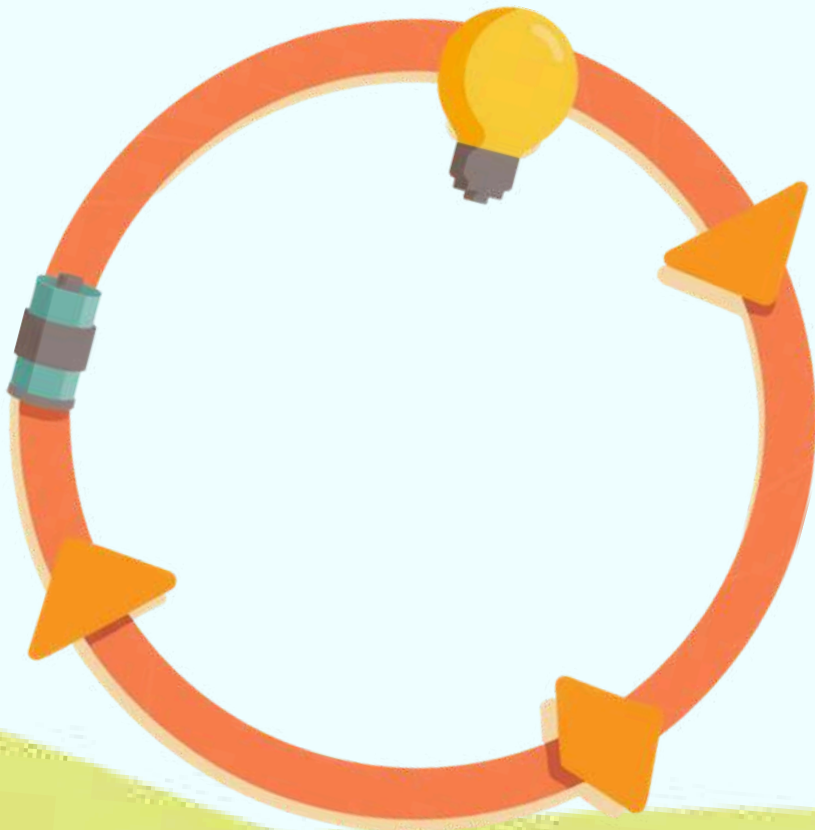
Conductor vs Insulator

What lets electricity travel through?



07

Electricity needs a complete path called a circuit. Think of it like a circle — energy travels all the way around!



Like Diwali garland bulbs — if one wire breaks, all the lights go off! The path must be complete for electricity to flow.

Circuit

Complete vs Broken Path



LED and Switch

Tiny lights that help us see!

LED is a tiny light bulb! You see them in street lights and fairy lights. It has a long leg (+) and a short leg (-). The long leg is the happy side!

A switch opens and closes the circuit. It is like your bedroom light switch! Press it ON and light glows. Press it OFF and light stops.



09

Fold your card in half like a greeting card. Press down on the fold to make it nice and flat. This will be your glowing card!

02

Stick copper tape inside your card to make a path. Press it down firmly so it sticks well. This is the road for electricity!

Building Our Circuit Card

Let's start making our glowing card!



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Step 3: Insert LED Legs

Push the long leg (+) of LED into one copper tape path. Push the short leg (-) into the other tape path. Make sure legs touch the tape!

04

Step 4: Place the Battery

Slide the coin cell battery under the copper tape. The shiny side (+) goes up! Press gently so tape touches the battery.



Building Our Circuit Card

Steps 3 & 4: Adding the LED and Battery



Tip 1

Draw with crayons around
your LED light

Tip 2

Add stickers to make it sparkle
and shine

Decorating the Card

Make your glowing card beautiful!

Use crayons to draw colorful pictures. Add fun
stickers like stars and flowers. Make your card
bright and happy. Your LED light will shine through
your art!



Test & Fix Your Circuit

If your light does not glow, don't worry!

- Check copper tape is stuck well
- Check LED legs are touching the tape
- Check battery is placed right side (+/-)
- Try pressing the battery gently



Question 1

What makes the light glow?

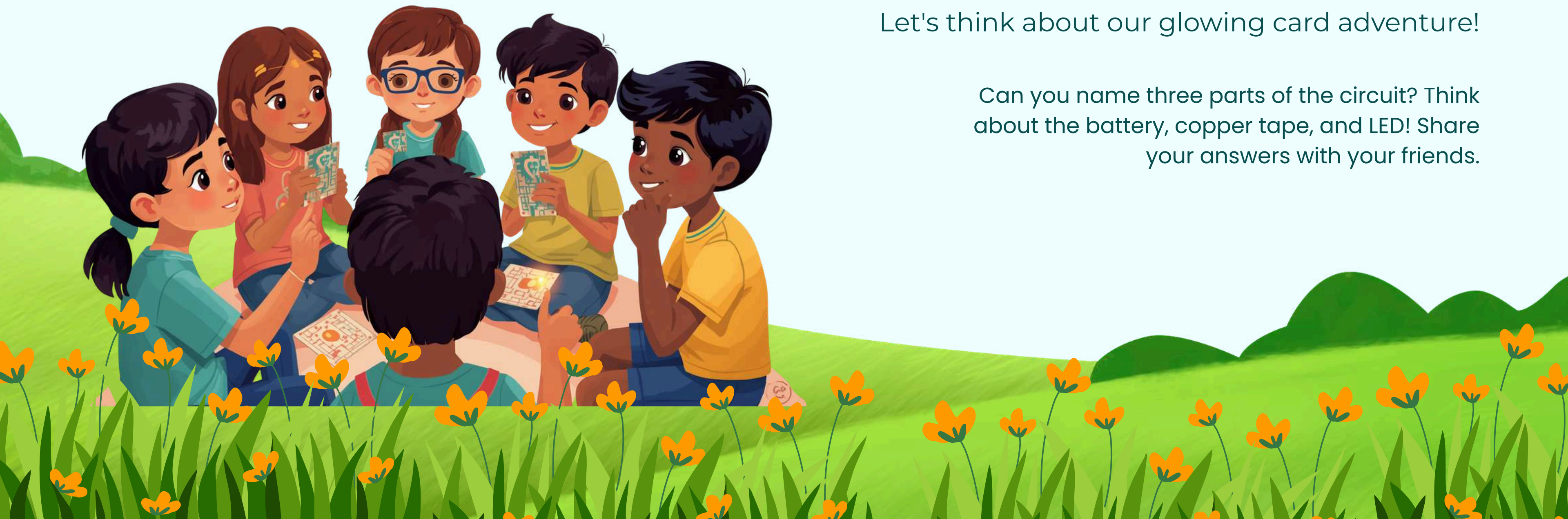
Question 2

What happens if battery lifts up?

Reflect: What Did We Learn?

Let's think about our glowing card adventure!

Can you name three parts of the circuit? Think about the battery, copper tape, and LED! Share your answers with your friends.



Add a second LED light to make your card extra bright and sparkly!

Extension Ideas

Try more fun circuit projects!

02

Draw a night scene with your LED as a glowing moon or twinkling star! Make a paper flap switch to turn your light on and off!





Congratulations! You Made a Glowing Card!

Well done! You made electricity work! Show
your glowing card with pride! Keep
exploring and creating!